TEXORA · LECTURE SHEET

Regulatory Issues: Environmental Policy and Law

The legal framework a textile factory operates under in Bangladesh, the clearance system built on the Environment Conservation Act and Rules, the roles of government and non-government bodies, international treaties, and hazard communication, penalties and compliance.

TE-4204 · ENVIRONMENTAL ENGINEERING AND POLLUTION CONTROL · TOPIC 8 · LEVEL 4 TERM 2

1. Why environmental law exists

Topics 1–7 described pollution as a technical problem; it is also a **market failure**. A factory that discharges untreated effluent shifts its treatment cost onto downstream fishers, farmers and water users who never agreed to bear it — an externality. Environmental policy exists to internalise that cost, and its instruments fall into three families: **command-and-control regulation** (standards, licences and clearances backed by penalties — the backbone in Bangladesh), **economic instruments** (discharge charges, tradable permits, subsidies for cleaner technology), and **voluntary/market instruments** (certification, buyer codes of conduct, eco-labelling). In the textile sector all three act at once: the law sets the floor, and international buyers' compliance requirements frequently demand performance above it.

2. Environmental law of Bangladesh

The framework rests on the Constitution's commitment to protection and improvement of the environment, and its operative statute is the **Bangladesh Environment Conservation Act, 1995** (with later amendments). The Act created the regulatory machinery: it empowers the government and the **Department of Environment (DoE)** to set standards, to require an **Environmental Clearance Certificate (ECC)** before an industrial unit is established or operated, to inspect premises, to direct corrective action, and to penalise offences — penalties under the Act include fines and imprisonment, with provision for closure of non-compliant operations. The **Environment Conservation Rules** made under the Act carry the working detail: ambient and discharge standards, the categorisation of industries, and the clearance procedure. Complementary laws bear on parts of factory life — factory

and labour law for worker safety, water and fisheries law for receiving waters — but the environmental clearance system is the gate through which every textile project passes.

2.1 Categorisation and the clearance process

Under the Rules, industries are classified by pollution potential into four categories — **Green, Yellow, Orange and Red** — in ascending order of scrutiny. Simple, low-pollution activities (Green) clear quickly with minimal documentation; the most polluting categories face full assessment. **Textile dyeing, washing and finishing plants fall in the Red category**: wet processing is, by law, treated as among the most pollution-intensive activities, and a Red-category project requires an **environmental impact assessment (EIA)** — a formal study of the site, the processes, the predicted impacts and the management plan (including the ETP design of Topic 6) — before clearance is granted.

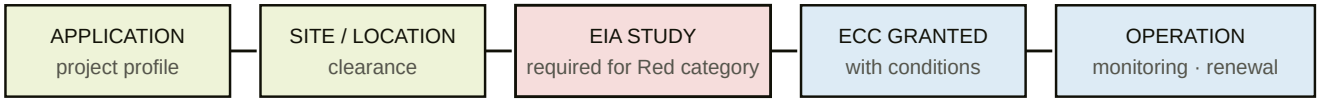


Figure 1 — The environmental clearance pathway for a Red-category textile project. The EIA gate is where the ETP and management plans of the earlier topics become legal commitments.

Clearance carries conditions: install and run the ETP, keep discharges within the standards for the category, maintain records, submit to inspection and monitoring, and renew the certificate periodically. Enforcement escalates from notices and directions to fines, disconnection of utilities in grave cases, and prosecution — and a factory may also face compensation claims for ecological damage. Foreign buyers audit against both the law and their own codes, so legal clearance is the minimum, not the target.

3. Roles of governmental and non-governmental organisations

BODY	ROLE IN ENVIRONMENTAL PROTECTION
Ministry of Environment, Forest and Climate Change	Policy, legislation, national strategy and international negotiation
Department of Environment (DoE)	The regulator: standards, clearance, EIA review, inspection, monitoring, enforcement
Local government and city authorities	Waste collection, drains, local planning and nuisance control

Standards and sector bodies (e.g. BSTI)	Product and process standards, testing
Industry associations (e.g. BGMEA, BKMEA)	Member guidance, compliance programmes, common facilities such as central ETPs in industrial estates
NGOs and environmental law groups	Research, public-interest litigation, community advocacy, watch-dog reporting (organisations such as the Bangladesh Environmental Lawyers Association are prominent examples)
Academic and research institutions	Monitoring data, technology development, EIA expertise, training
International buyers and certification schemes	Market enforcement: audits, chemical and wastewater requirements, certification as a condition of orders

The system works when the parts interlock: policy sets direction, the DoE regulates, associations spread capability among factories, NGOs and courts supply public pressure and legal recourse, and buyers make compliance commercially necessary. Its known weaknesses — enforcement capacity, monitoring coverage, and the gap between paper clearance and daily practice — are precisely the gaps a competent factory environmental manager closes inside his own plant.

4. International treaties and conventions

TREATY	SUBJECT	TEXTILE RELEVANCE
Montreal Protocol	Phase-out of ozone-depleting substances (Topic 2)	Refrigerants, foam and aerosol choices in the factory
Stockholm Convention	Persistent organic pollutants (POPs)	Certain flame retardants, water/oil repellents and legacy pesticides on fibre
Basel Convention	Transboundary movement of hazardous waste	Export/recovery shipments of sludge, waste chemicals and contaminated materials
Rotterdam Convention	Prior informed consent for hazardous chemicals in trade	Import of listed dyes and process chemicals
UNFCCC and Paris Agreement	Climate change: national commitments, reporting	Energy and emissions data buyers request from suppliers
Ramsar Convention	Wetlands of international importance	Discharge discipline near listed wetlands (Bangladesh sites include Tanguar Haor and the Sundarbans)

Alongside treaties, the **health and industrial safety** framework — labour law domestically and ILO conventions internationally — governs machine safety, chemical exposure, fire and building safety, areas where the Bangladesh garment industry's compliance systems were substantially rebuilt after the disasters of the early 2010s. And for dyes and pigments specifically, market law matters as much as environmental law: the EU's REACH system and equivalent regimes restrict hazardous substances in imported articles, so **regulatory affairs for dyes and pigments** — knowing which substances are banned or restricted in each market and proving it by testing — is a standing function in any export factory.

5. Risk versus hazard communication

Two words run through all chemical regulation and must not be conflated. A **hazard** is an intrinsic property — a chemical's capacity to burn, corrode, poison or sensitise. **Risk** is the probability and severity of harm in real use, which depends on exposure:

$$\text{Risk} \approx \text{Hazard} \times \text{Exposure}$$

A highly hazardous chemical used in a closed, automated system may present low risk; a mildly hazardous one sprayed openly by hand all shift may present high risk. **Hazard communication** is the system that makes hazards known wherever chemicals travel: the **Globally Harmonized System (GHS)** gives standard label elements (pictograms such as flame, corrosion, exclamation mark, skull-and-crossbones, health hazard and environment), signal words ("Danger" / "Warning"), hazard and precautionary statements, and the 16-section **safety data sheet (SDS)** that must accompany chemicals into the store and the dyehouse. Risk communication goes further — it tells workers and communities what the actual risk is, what controls are in place, and what to do in an incident — and it fails when it either hides hazards or cries wolf about negligible risks.

6. Penalties and compliance

Non-compliance is punished on several tracks at once. **Regulatory:** directions to correct, fines and, for continuing or serious offences under the Environment Conservation Act, imprisonment and closure; compensation for ecological damage may also be ordered. **Commercial:** loss of clearance status, failed buyer audits, suspended orders and removal from approved-supplier lists — for an export factory often the heavier penalty. **Civil and reputational:** claims by affected communities and public exposure. A compliance programme therefore treats the topics of this whole course as one management system: source control (Topic 3), treatment (Topics 4–7), monitoring records against the standards (Topics 4–6), trained people, honest reporting, and periodic internal audit — the factory that runs its ETP daily, keeps its analysis log and knows its numbers is the factory that passes inspections it did not schedule.

Glossary

Environment Conservation Act, 1995 — Bangladesh's principal environmental statute. **DoE** — Department of Environment, the regulator. **ECC** — Environmental Clearance Certificate, required before establishing/operating an industrial unit. **EIA** — environmental impact assessment, required for Red-category projects. **Green/Yellow/Orange/Red categories** — industry classification by pollution potential under the Environment Conservation Rules. **Hazard** — intrinsic capacity of a substance or situation to cause harm. **Risk** — likelihood × severity of harm in actual exposure. **GHS** — Globally Harmonized System of classification and labelling of chemicals. **SDS** — safety data sheet. **REACH** — the EU's chemicals regulation, restricting hazardous substances in products including textiles. **Externality** — a cost of production imposed on third parties outside the price mechanism.

Key points

• Pollution is an externality; policy answers with regulation, economic instruments and market pressure together.

• Bangladesh's framework: Environment Conservation Act 1995 → DoE → categorisation (Green/Yellow/Orange/Red) → clearance; dyeing and washing are Red category and require an EIA.

• Government sets and enforces; associations build capability; NGOs and courts add public pressure; buyers make compliance commercial.

• Treaties touching textiles include Montreal, Stockholm, Basel, Rotterdam, the climate conventions and Ramsar; REACH governs chemicals in export markets.

• Risk = hazard × exposure; GHS labels and SDS communicate hazards; penalties for non-compliance are regulatory, commercial and civil at once.

Practice

1. Sketch the clearance pathway for a new knit dyeing factory in Bangladesh, naming the category it falls in and the study that category requires.
2. Distinguish hazard from risk with one chemical example of high hazard/low risk and one of low hazard/high risk in a dyehouse.
3. Match each treaty to its subject: Montreal, Stockholm, Basel, Ramsar — and state one concrete implication of each for a textile factory.
4. List the environmental roles of (a) the DoE, (b) an industry association, (c) an environmental NGO, and say how their actions interlock.
5. A factory's ETP is bypassed at night to save chemical costs. Enumerate the penalty tracks it is exposed to, beyond the discharge offence itself.

Answer key

1. Application with project profile → site/location clearance → EIA study and its approval → ECC granted with conditions → operation with monitoring and periodic renewal. Knit dyeing is a **Red-category** industry; the study required is a full **environmental impact assessment (EIA)**, including the ETP design.
2. High hazard/low risk: a concentrated corrosive alkali stored in a bunded tank farm and dosed by closed automatic lines — severe intrinsic hazard, minimal exposure. Low hazard/high risk: a low-toxicity dye dust or enzyme powder handled openly by hand all shift, inhaled daily — modest intrinsic hazard, continuous exposure makes real risk. (Any chemically sensible pair accepted.)
3. Montreal — ozone-depleting substances; implication: compliant refrigerants/foam agents only. Stockholm — persistent organic pollutants; implication: certain repellent/flame-retardant chemistries barred from recipes. Basel — transboundary hazardous waste; implication: sludge/waste-chemical shipments across borders need consent and documentation. Ramsar — listed wetlands; implication: strict discharge discipline for any plant draining toward Tanguar Haor or the Sundarbans catchments.
4. (a) DoE: sets standards, issues clearance, reviews EIAs, inspects and enforces. (b) Association: guides members on compliance, organises training and common facilities (e.g. central ETPs), represents the sector. (c) NGO: research, awareness, public-interest litigation and watch-dog reporting. Interlock: the regulator's rules are the floor, associations raise factories' capability to meet them, NGOs and courts police the gap between paper and practice — with buyers' audits reinforcing all three.
5. Regulatory: directions, fines and possible imprisonment/closure under the Environment Conservation Act, plus ecological-damage compensation; licence consequences up to loss or non-renewal of the ECC. Commercial: failed buyer audits, suspended or cancelled orders, removal from approved-supplier lists. Civil/reputational: claims by affected downstream users and communities, media exposure, and loss of certification. The night-time bypass also falsifies monitoring records, compounding the offence.
